

XRPL Validator — Daily Engineering Report • 2026-10-09

Generated 2026-10-10 00:21 EDT. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the live native-engine shadow, the reference rippled node, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **8 commits** (xrpl-validator 8)
- **1 engine finding fixed** (F417)
- Live shadow: 10,840/10,840 ledgers fully matched, 986,414 transactions, 0 engine receipts in the current soak
- Amendment **fixCleanup3_4_0** gained majority at ledger 107,536,129 (2026-10-09 09:34 UTC)
- Amendment **fixBatchV1_2** was enabled at ledger 107,540,481 (2026-10-09 14:15 UTC)
- Amendment **BatchV1_1** was enabled at ledger 107,540,993 (2026-10-09 14:47 UTC)
- Amendment **fixCleanup3_4_0** held majority (since 2026-10-09 09:34 UTC); activation due ~2026-10-23 09:34 UTC

Live validator and native engine

Snapshot at generation time.

Measure	Value
Ledgers compared (current run)	10,840
Ledgers fully matched	10,840
Transactions applied natively	986,414
Result mismatches	0
Byte mismatches	0
Missing / extra keys	0 / 0
Batch inner-result mismatches	0
Engine receipts (current soak)	0
Live engine build running since	2026-10-09 12:39 EDT
Reference rippled	3.4.1, full, validated #107,553,638

Engine findings fixed

- **fix(node): F417 — a Batch inner is filed under the outer its ParentBatchID names (Mainnet #107541023 FAD38FDE4D09)** (b29d83a , xrpl-validator) The first mainnet Batches after BatchV1_1 (2026-10-09) came from one bot: up to eight tfAllOrNothing outers from different accounts per ledger, every one wrapping the SAME ticketed pair of inner payments. rippled discards the outers whose round trip falls short of DeliverMin and files the pair under the first one that clears it; the rest get tefNO_TICKET. The engine applied every outer exactly so. But the inner ...

Engineering changes

- **fix(probe): differential_probe keeps a tecEXPIRED's erasures, as the node does (devnet #5533675 7E7F5D06640B)** (59a0ddc , xrpl-validator) The probe applies through its own copy of native_apply_one, and that copy restored the post-common snapshot on every claimed tec but tecKILLED. The node's copy settles a tecEXPIRED instead (finding 245: apply::settle_expired keeps the NFTTokenOffer / Credential erasures and their directory and OwnerCount moves). So a Payment naming an expired credential (campaign 23, devnet #5533675) kept the credential the network ...
- **fix(ffi): leg A credits a shared Batch inner to the outer its ParentBatchID names, not the last carrier** (863651f , xrpl-validator) F417's other half. Several outers can carry the very same inner and only one files it (mainnet #107541023: four outers wrapping one ticketed pair, the first discarded, the third filing the pair, the fourth after it). The native side pairs inner verdicts by ParentBatchID since b29d83a; leg A (ffi_engine's C++ compare) still filled inner_of in ledger order, so a shared inner's expected mutations folded into the LAST ...
- **fix(flow): rippled 3.4.1's overflow hardening — checked savedIns/savedOuts sums in flow() and BookStep** (3a9c669 , xrpl-validator) rippled 3.4.1 (unconditional, no amendment) folds every strand-level and book-level aggregate through checkedStepAddOpt: XRP (and MPT) add in checked int64, IOU through Number as before.
- **fix(ledger): fixBatchV1_2 — a Batch inner must be a RawTransaction object (rippled 3.4.1)** (5710fd4 , xrpl-validator) rippled 3.4.1's one gated change (Batch.cpp:293): with fixBatchV1_2, every RawTransactions entry must be wrapped as RawTransaction, or the batch is temMALFORMED. The check is FIRST in the per-inner loop, ahead of the uniqueness check. fixBatchV1_2 was ON on mainnet by #107540484 (2026-10-09), ahead of BatchV1_1, so no mainnet ledger ran Batch without it; devnet and testnet did.

Docs and operations

- **docs: the plan to a fully independent validator, updated for Stage 4 Phase B** (66081c8 , xrpl-validator)
- **docs: the independence plan, in depth** (7cf3aff , xrpl-validator)
- **docs: the independence task breakdown and the recommended order** (74a981e , xrpl-validator)

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Voting events:

- fixCleanup3_4_0 gained majority — ledger 107,536,129, 2026-10-09 09:34 UTC
- fixBatchV1_2 was enabled — ledger 107,540,481, 2026-10-09 14:15 UTC
- BatchV1_1 was enabled — ledger 107,540,993, 2026-10-09 14:47 UTC

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-09 09:34 UTC	2026-10-23 09:34 UTC

Amendments enabled on mainnet: 97.

Appendix — commits

Time	Repo	Commit	Subject
11:26	xrpl-validator	b29d83a	fix(node): F417 — a Batch inner is filed under the outer its ParentBatchID names (Mainnet #107541023 FAD38FDE4D09)
12:09	xrpl-validator	59a0ddc	fix(probe): differential_probe keeps a tecEXPIRED's erasures, as the node does (devnet #5533675 7E7F5D06640B)
18:46	xrpl-validator	66081c8	docs: the plan to a fully independent validator, updated for Stage 4 Phase B
18:51	xrpl-validator	7cf3aff	docs: the independence plan, in depth
19:18	xrpl-validator	74a981e	docs: the independence task breakdown and the recommended order
22:42	xrpl-validator	863651f	fix(ffl): leg A credits a shared Batch inner to the outer its ParentBatchID names, not the last carrier
22:42	xrpl-validator	3a9c669	fix(flow): rippled 3.4.1's overflow hardening — checked savedIns/savedOuts sums in flow() and BookStep
22:42	xrpl-validator	5710fd4	fix(ledger): fixBatchV1_2 — a Batch inner must be a RawTransaction object (rippled 3.4.1)